

REAL-TIME SIGN LANGUAGE INTERPRETER

Ayushi Singh
24WH1A0554

Sajja Lakshmi Rajyam
24WH1A0501

Under the Guidance of
Dr. Aruna Rao S L
Head of the Department
Dept of CSE
BVRIT Hyderabad College of Engineering for Women

April 23, 2026

- ➊ Abstract
- ➋ Introduction
- ➌ Objectives
- ➍ Technology Stack
- ➎ System Design
- ➏ Implementation
- ➐ Results
- ➑ Output
- ➒ Conclusion
- ➓ Future Enhancements
- ➔ Thank You

This project presents a real-time Sign Language Interpreter that translates ASL gestures into text and speech.

It uses:

- Computer Vision (MediaPipe)
- Machine Learning (Random Forest)
- Real-time webcam processing

The system recognizes 26 alphabets with high accuracy and enables seamless communication between deaf and hearing individuals.

- Around 63 million deaf people in India face communication barriers
- Sign language is not widely understood
- Existing systems are costly or require special hardware

Solution: A real-time, affordable interpreter using a standard webcam.

- Detect hand landmarks using MediaPipe
- Recognize ASL alphabets (A–Z)
- Provide real-time processing (16 FPS)
- Convert gestures into text
- Enable text-to-speech output
- Build an intuitive user interface

Languages & Libraries

- Python
- OpenCV
- MediaPipe
- NumPy, Pandas
- Scikit-learn

Tools

- Streamlit / Flask (UI)
- pyttsx3 (Text-to-Speech)

- Video Capture via webcam
- Hand Detection using MediaPipe
- Feature Extraction (21 landmarks)
- Classification using Random Forest
- Text Buffer + Speech Output

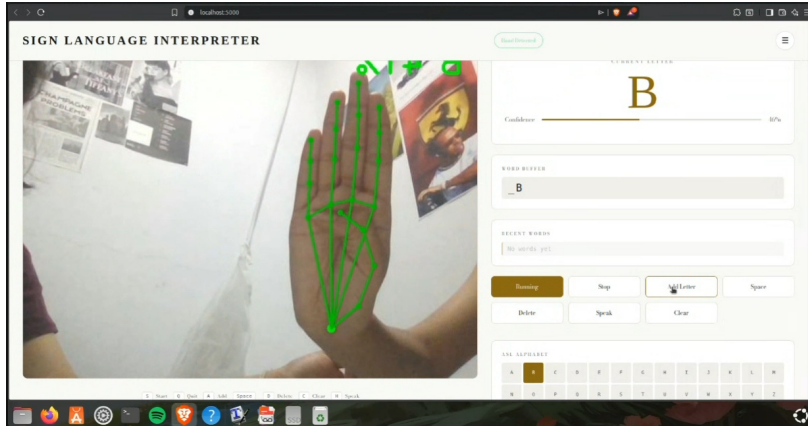
- ① Capture frame
- ② Detect hand landmarks
- ③ Normalize features
- ④ Predict gesture
- ⑤ Check stability
- ⑥ Display text / generate speech

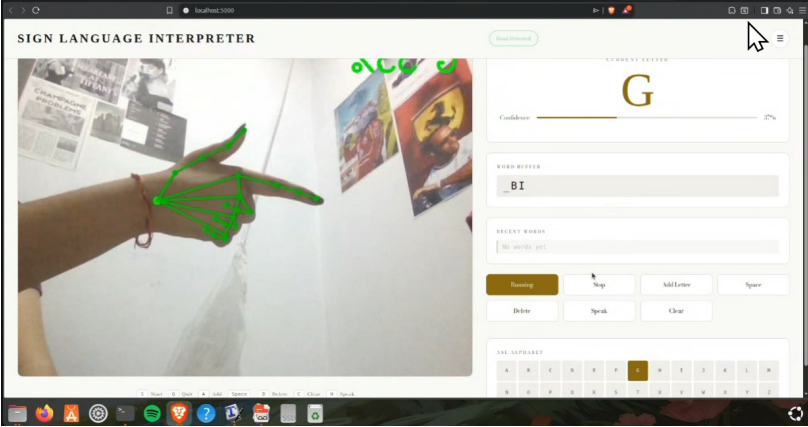
- 21 landmarks extracted per hand
- 42 features used for classification
- Random Forest model (200 trees)
- Accuracy: 81%
- Stability check: 16 frames

- Accuracy: 81%
- Frame Rate: 16-18 FPS
- Latency: 53 ms
- Works on CPU (no GPU required)

Outcome: Efficient real-time gesture recognition system.







- Successfully developed real-time interpreter
- Works on standard hardware
- Provides both text and speech output
- Bridges communication gap effectively

- Word-level recognition using LSTM
- Multi-hand gesture support
- Cloud deployment
- Mobile application development

Thank You